



E.2 Quantum physics (HL)

Nuclear and quantum physics · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Light exhibits particle-like behaviour, demonstrated by the photoelectric effect: electrons are emitted from a metal surface only if incident photons have frequency above a threshold f_0 (dependent on the metal's work function $\phi = hf_0$), with maximum kinetic energy of emitted electrons given by $E_{k(\max)} = hf - \phi$, independent of light intensity, contradicting classical wave predictions. Matter exhibits wave-like behaviour (wave-particle duality), with the de Broglie wavelength of a particle given by $\lambda = h/p = h/(mv)$, confirmed by electron diffraction experiments. The Heisenberg uncertainty principle states that certain pairs of conjugate quantities cannot both be known to arbitrary precision simultaneously, notably $\Delta x \cdot \Delta p \geq h/4\pi$ for position and momentum, and $\Delta E \cdot \Delta t \geq h/4\pi$ for energy and time, reflecting a fundamental limit rather than measurement error.

Worked example: follow the reasoning

Light of frequency 7.5×10^{14} Hz shines on a metal with work function 2.0 eV. Find the maximum kinetic energy of emitted photoelectrons in eV. ($h = 6.63 \times 10^{-34}$ J·s, $1 \text{ eV} = 1.60 \times 10^{-19}$ J)
 $E_{\text{photon}} = hf = (6.63 \times 10^{-34})(7.5 \times 10^{14}) = 4.973 \times 10^{-19} \text{ J} = 4.973 \times 10^{-19} / 1.60 \times 10^{-19} = 3.11 \text{ eV}$
 $E_{k(\max)} = E_{\text{photon}} - \phi = 3.11 - 2.0 = 1.11 \text{ eV}$
 Answer: $E_{k(\max)} \approx 1.1 \text{ eV}$.

Quick check · multiple choice

In the photoelectric effect, increasing the intensity of incident light (above the threshold frequency) while keeping frequency constant causes:
 A An increase in the maximum kinetic energy of photoelectrons
 B An increase in the number of photoelectrons emitted per second, but no change in their maximum kinetic energy
 C A decrease in the threshold frequency
 D No photoelectric effect at all

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An electron is accelerated from rest through a potential difference of 150 V. ($h = 6.63 \times 10^{-34}$ J·s, $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C) [7]
 (a) Calculate the kinetic energy gained by the electron in joules.
 (b) Calculate the speed of the electron.
 (c) Calculate the de Broglie wavelength associated with the electron.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $E_k = eV = (1.60 \times 10^{-19})(150) = 2.40 \times 10^{-17} \text{ J}$.

(b) $\frac{1}{2}mv^2 = E_k \rightarrow v = \sqrt{2E_k/m} = \sqrt{2(2.40 \times 10^{-17})/9.11 \times 10^{-31}} = \sqrt{5.269 \times 10^{13}} = 7.26 \times 10^6 \text{ m}\cdot\text{s}^{-1}$.

(c) $\lambda = h/(mv) = (6.63 \times 10^{-34})/[(9.11 \times 10^{-31})(7.26 \times 10^6)] = 6.63 \times 10^{-34}/6.616 \times 10^{-24} = 1.00 \times 10^{-10} \text{ m}$.

Quick check answer

B. Intensity corresponds to the number of photons per second, not their individual energy; more photons release more electrons per second, but each photon still transfers the same energy hf , so maximum kinetic energy ($hf - \phi$) is unchanged.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.